

MorphMaster User Manual

MorphMaster is intended for users who are already comfortable with multi-track MIDI configuration on their instrument, including concepts such as control tracks, Local Off (or an equivalent local-control setting), and bank/program selection. The sections that follow assume this background. If any of these terms are unfamiliar, consult your instrument's MIDI documentation on multi-part setup and local control before continuing.

MorphMaster is a MIDI performance tool that receives performance data from a single MIDI control track and uses it to control up to sixteen MIDI receive tracks on the selected MIDI output device. The sound produced by each track can be continuously morphed according to:

- Keyboard position (low notes → high notes)
- Playing velocity (soft → loud)
- A combination of both parameters

MorphMaster is designed to create expressive layered sounds that evolve naturally across the keyboard and dynamic range.

Main Concepts

MorphMaster receives MIDI notes from a single MIDI input device and distributes them to multiple virtual tracks.

Each track can:

- Use a different instrument sound
- Occupy a different region of the morphing surface
- Respond differently to note position and velocity
- Apply its own volume, pan and timbre settings

The final sound is obtained by mixing the contribution of all active tracks.

Before You Start

Before using MorphMaster, the MIDI input and output settings of the remote instrument must be configured on the remote instrument itself. These settings cannot be made from within MorphMaster.

The remote instrument must provide at least one MIDI track that sends data out of the instrument and into MorphMaster. This track is the **control track**.

The control track must be different from the tracks that MorphMaster will control. Its purpose is only to send MIDI events to MorphMaster, not to produce the final sound.

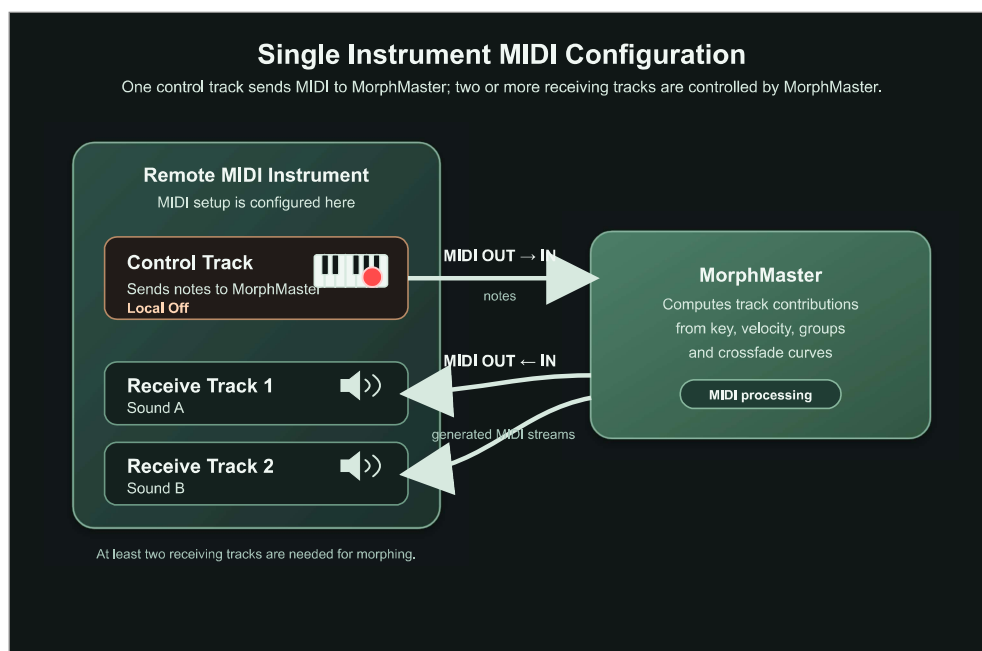
The remote instrument must also provide at least two MIDI tracks that receive data from MorphMaster. These are the tracks whose sounds will be blended by MorphMaster.

At least two receiving tracks are required because morphing is based on a gradual transition between two or more sounds. With only one receiving track, there would be nothing to morph between.

The control track should be set to **Local Off**, or to an equivalent setting provided by the remote instrument.

This means that the control track should generate MIDI events only on the instrument's MIDI output, without directly playing the internal sound engine of the instrument.

If the control track also plays the internal sound engine, its sound will be mixed with the MIDI stream coming back from MorphMaster. This can mask the morphing effect and make the transitions between sounds less clear.

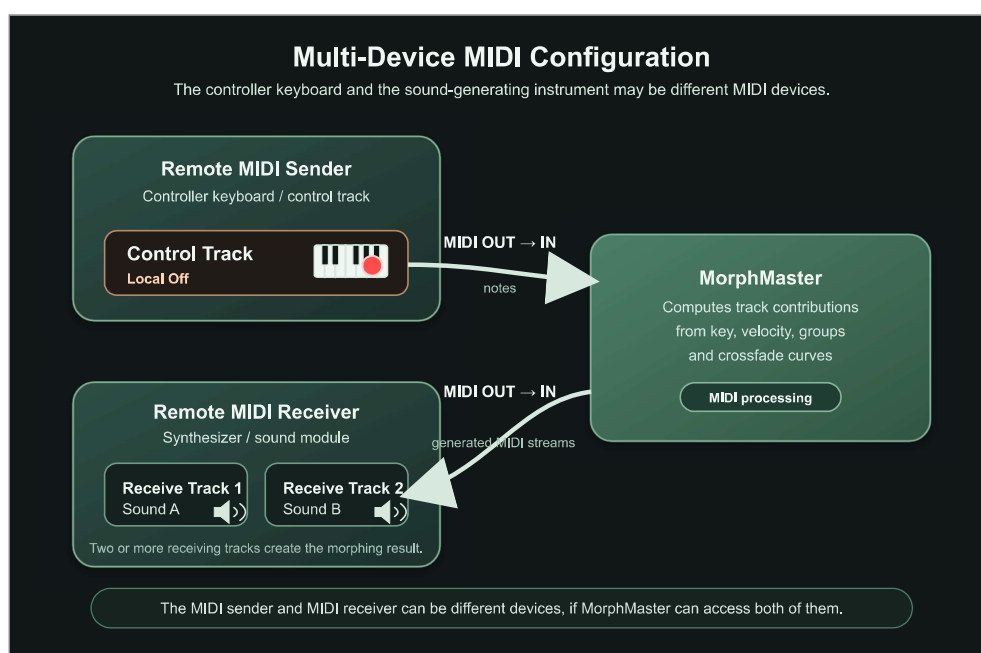


MorphMaster configuration using a single MIDI instrument.

The control track and the receive tracks do not necessarily need to belong to the same instrument.

If the computer running MorphMaster can access multiple MIDI devices, one instrument may be used as the playing controller while a different instrument (or sound module) may be used to receive the MIDI data generated by MorphMaster.

This configuration can be useful, for example, when a master keyboard is used to play sounds generated by a separate synthesizer, workstation or sound module.



MorphMaster configuration using separate MIDI sender and receiver devices.

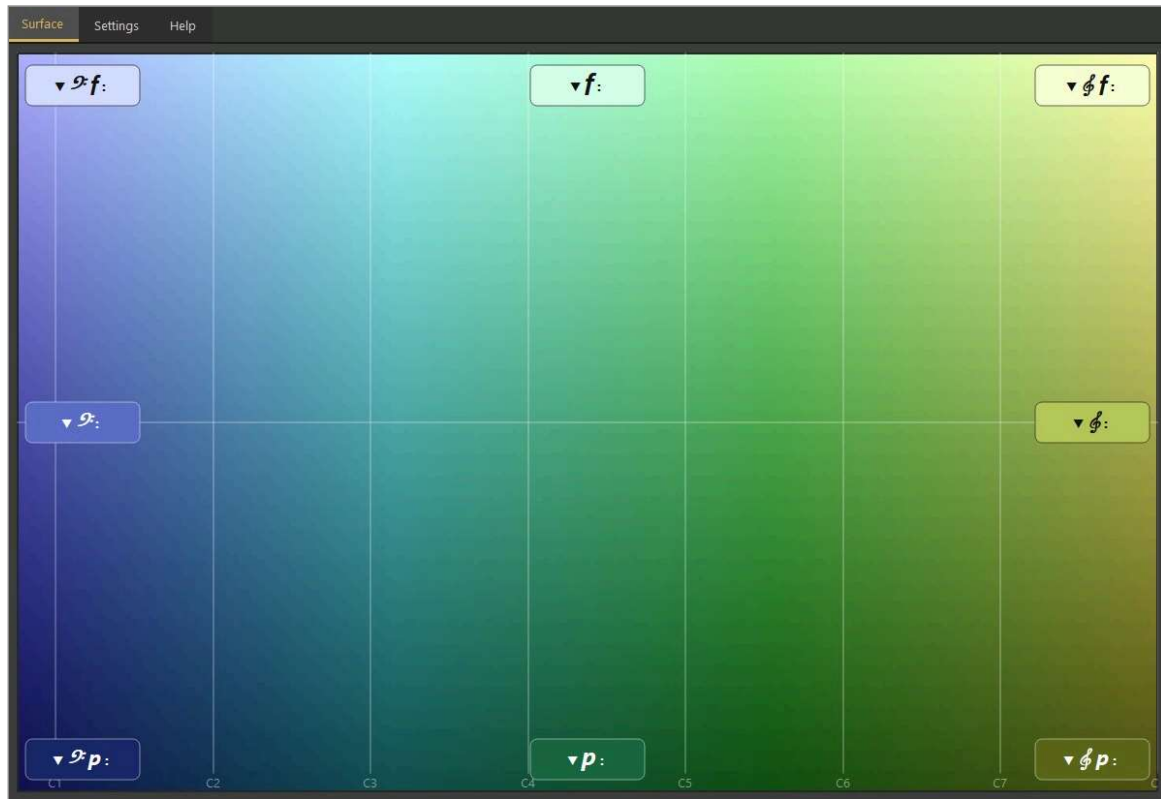
User Interface Overview

The application is divided into three main pages:

- **Surface** – Real-time performance and morphing visualization
 - **Settings** – MIDI configuration and preset management
 - **Help** – Documentation and application information
-

1. Surface Page

Surface Page



Main morphing surface.

The surface represents the two dimensions used for morphing:

- Horizontal axis = keyboard position
- Vertical axis = velocity

The horizontal axis is scaled according to the **Remote Keyboard Range** selected in **Settings** → **General**. This allows the surface to represent the actual note range of the connected keyboard, avoiding unused areas when the remote keyboard has fewer than 128 possible MIDI notes.

Moving from left to right changes the contribution of the tracks assigned to the horizontal groups.

Moving from bottom to top changes the contribution of the tracks assigned to the vertical groups.

Morphing Groups

MorphMaster uses eight morphing groups positioned around the performance surface. The horizontal axis represents keyboard position (Bass ↔ Treble), while the vertical axis represents playing dynamics (Piano ↔ Forte). The corner groups combine both dimensions.

Badge	Group	Description
♭:	Bass	Low keyboard range
♭:f	Bass Forte	Low keyboard range, loud dynamics
f	Forte	Loud dynamics
♭f	Treble Forte	High keyboard range, loud dynamics
♭	Treble	High keyboard range
♭p	Treble Piano	High keyboard range, soft dynamics
p	Piano	Soft dynamics
♭p	Bass Piano	Low keyboard range, soft dynamics

The badges displayed around the surface indicate which tracks are currently assigned to each group.

Example

For example, **f : 4** means that Track 4 belongs to the **Forte** group, while **p : 3** means that Track 3 belongs to the **Piano** group.

Tracks assigned to the **Piano** group contribute more strongly when notes are played softly, with low velocity. Conversely, tracks assigned to the **Forte** group contribute more strongly when notes are played loudly, with high velocity.

In this example, when playing very softly, the sound is dominated by the contribution of Track 3. As playing intensity increases, the contribution of Track 3 gradually decreases while the contribution of Track 4 gradually increases.

The transition is continuous rather than abrupt, creating a smooth morph between the two sounds as velocity changes. The exact shape of this transition depends on the morphing curves assigned to the tracks. Morphing curves are described later in this manual.

The same principle applies to keyboard position.

In the example shown on the Morphing Surface, Tracks 1 and 2 belong to the **Bass** (♭:) group, while Track 6 belongs to the **Treble** (♭) group.

Notes played in the lower part of the keyboard favour Tracks 1 and 2. As notes move towards the upper part of the keyboard, the contribution of the Bass tracks gradually decreases while the contribution of Track 6 gradually increases.

At intermediate keyboard positions, both Bass and Treble tracks may contribute simultaneously, producing a smooth transition. The relative contribution of each track is determined by the assigned morphing curves.

Since keyboard position and velocity are evaluated simultaneously, MorphMaster can blend sounds across both dimensions at the same time.

For example, a soft note played in the lower register will favour the **Bass Piano** ($\text{B}\flat\text{p}$) region, while a loud note played in the upper register will favour the **Treble Forte** (Tf) region.

Notes played between these extremes may activate several groups simultaneously, creating rich and continuously evolving timbral transitions. The exact balance between the active groups depends on the morphing curves assigned to the corresponding tracks.

Assigning Tracks to Groups



To assign tracks to a group, touch or click the badge of the desired group on the Morphing Surface.

A menu opens showing the available tracks. Select the tracks that should belong to that group.

Once assigned, the selected tracks appear inside the badge. For example, **f : 5, 6** means that Tracks 5 and 6 are assigned to the **Forte** group.

Track assignment menu opened from a group badge.

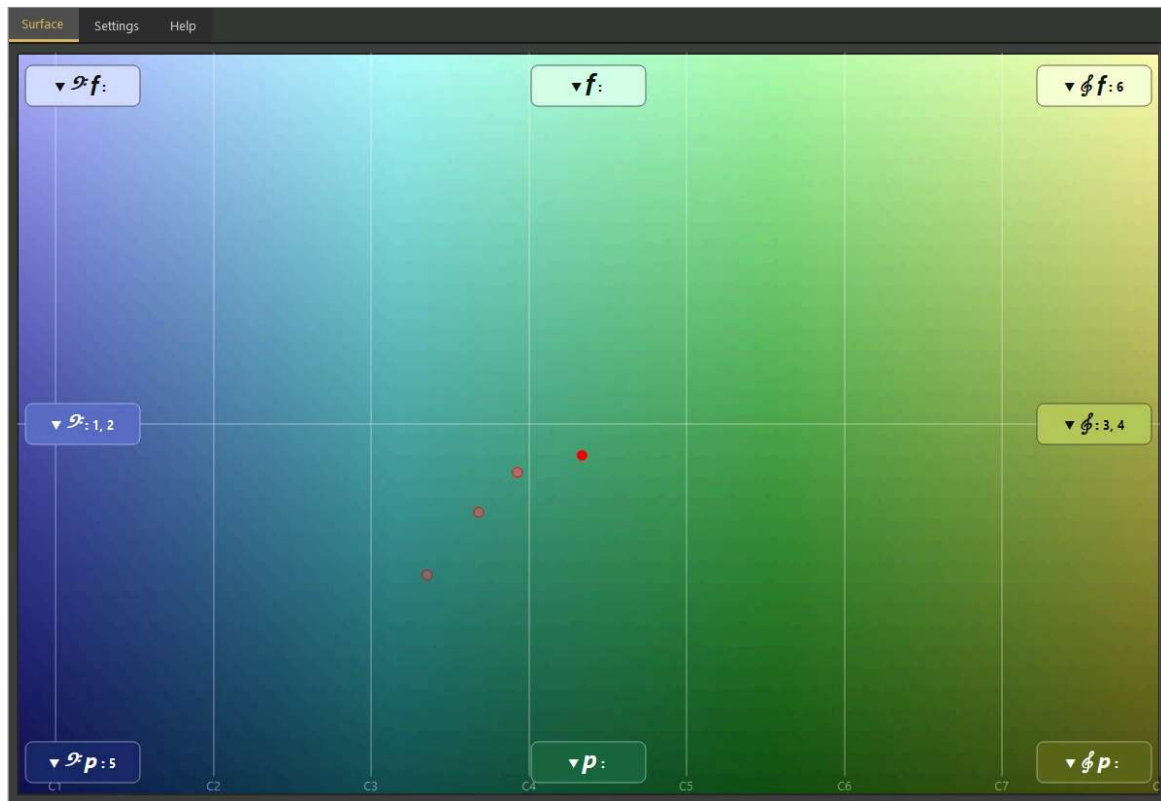
Real-Time Note Display

Notes played on the remote MIDI keyboard are displayed on the Morphing Surface as red dots.

Each dot represents one played note. Its position corresponds to the point with coordinates (**key, velocity**):

The key coordinate is interpreted within the currently selected **Remote Keyboard Range**.

- The horizontal position represents the played key, from low notes on the left to high notes on the right.
- The vertical position represents velocity, from soft notes at the bottom to loud notes at the top.



Red dots show the notes currently played on the MIDI keyboard, positioned according to key and velocity.

These note positions are used by MorphMaster to calculate the contribution of the tracks assigned to the different morphing groups.

Interactive Surface

The Morphing Surface is not only a visual representation of incoming MIDI notes. It is also an active playing surface.

Touching or clicking any point on the surface generates a note whose keyboard position and velocity are derived directly from the selected location:

- The horizontal coordinate determines the note position (key), within the currently selected Remote Keyboard Range.
- The vertical coordinate determines the note velocity.

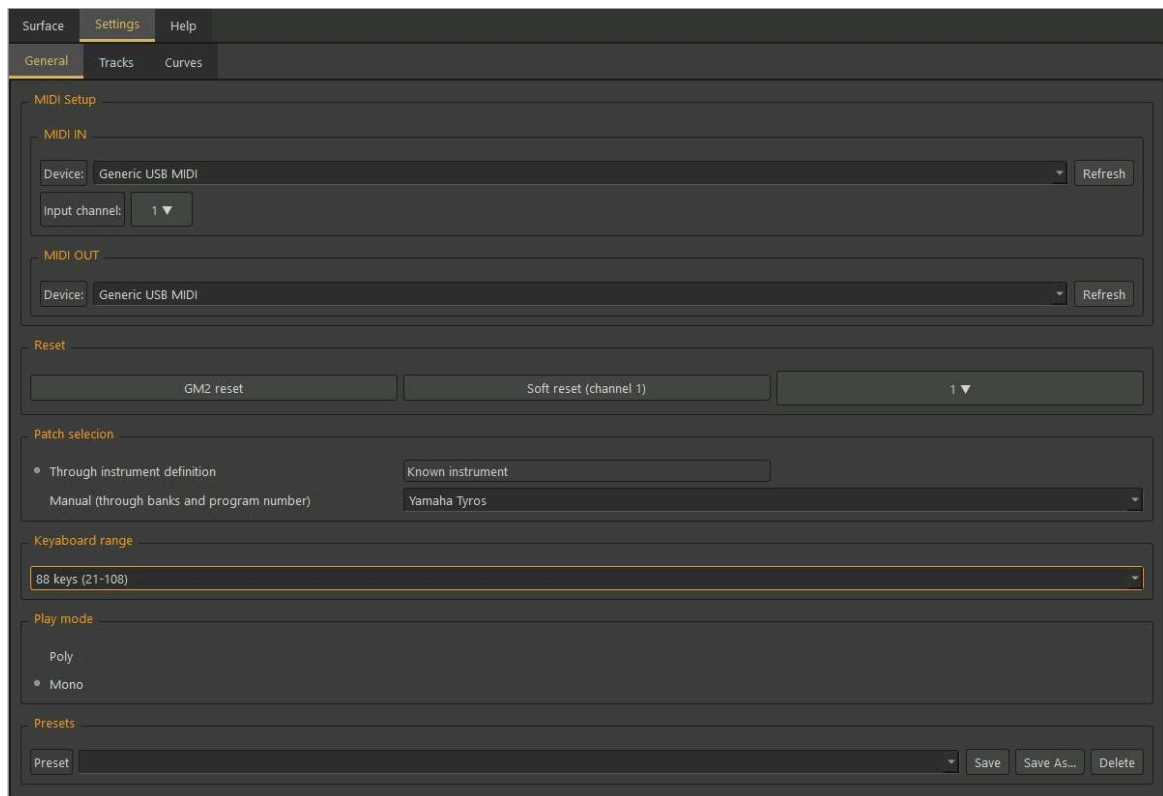
The generated note is then processed exactly like a note received from the connected MIDI keyboard. Its contribution to the final sound depends on the groups to which the tracks are assigned and on the morphing curves associated with those groups.

The details of the morphing curves are explained later in this manual.

While the surface is not intended to replace a musical keyboard for performing melodies, it provides a fast and intuitive way to explore the behaviour of a preset.

It is particularly useful when assigning tracks to groups, adjusting morphing curves, or experimenting with different morphing strategies, because the resulting timbral changes can be heard immediately without the need to play a specific sequence on a MIDI keyboard.

2. Settings → General



General settings.

MIDI Setup

MIDI Input

Select the MIDI device that provides the incoming performance data to MorphMaster.

In a typical configuration, this is the device that contains the **control track** described in the **Before You Start** section. The control track sends note events and velocity information to MorphMaster, which uses them to calculate the contribution of the various tracks assigned to the morphing groups.

The selected device may be the same instrument that will later receive the MIDI data generated by MorphMaster, or it may be a completely different device acting only as a MIDI controller.

The **Input Channel** selector specifies the MIDI channel used by the control track. MorphMaster will listen only to messages received on the selected channel.

If MIDI devices are connected or disconnected while MorphMaster is running, press the **Refresh** button to update the list of available MIDI input devices.

MIDI Output

Select the MIDI device that will receive the MIDI messages generated by MorphMaster.

In a typical configuration, this is the instrument, synthesizer or sound module that contains the **receive tracks**. MorphMaster sends MIDI data to these tracks and continuously adjusts their contribution according to keyboard position, playing velocity, group assignments and morphing curves.

The selected output device may be the same instrument used as the MIDI input source, or it may be a separate MIDI receiver dedicated to sound generation.

If MIDI devices are connected or disconnected while MorphMaster is running, press the **Refresh** button to update the list of available MIDI output devices.

Reset Section

GM2 Reset

Sends a General MIDI 2 reset message to the output device.

Soft Reset

Resets the selected MIDI channel without performing a complete GM2 reset.

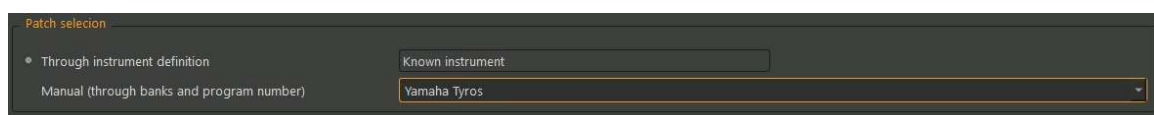
Patch Selection

MorphMaster supports two different methods of selecting sounds for each track.

Through Instrument Definition

This is the recommended method when the target instrument is included in MorphMaster's instrument database.

First, select the instrument that you want MorphMaster to control. The instrument database contains definitions for a number of supported MIDI instruments and sound modules.



Selected instrument.

Once an instrument has been selected, sounds can be browsed using two additional selectors:

- A **Category** selector.
- A **Program** selector.

The available categories are:

- Piano
- Electric Piano
- Chromatic Percussion
- Accordion
- Organ
- Guitar
- Bass
- Strings

- Ensemble
- Brass
- Reed
- Pipe
- Synth Lead
- Synth Pad
- Synth Effects
- Ethnic
- Percussive
- Sound Effects
- Other

After selecting a category, the Program selector displays all sounds belonging to that category for the selected instrument. Program names appear exactly as they are defined for the target instrument, making it easy to locate and select sounds without referring to MIDI bank and program numbers.

When a sound is selected, MorphMaster automatically sends the appropriate Bank Select and Program Change messages required by the target instrument.

Manual Mode

Manual Mode allows direct entry of MIDI patch information:

- Bank MSB
- Bank LSB
- Program Number

This mode is useful when controlling devices that are not included in MorphMaster's instrument database, or when a specific sound must be selected using its raw MIDI bank and program numbers.

Remote Keyboard Range

The **Remote Keyboard Range** setting specifies the effective note range of the keyboard used as the MIDI sender.

This setting is useful when the remote keyboard does not cover the full MIDI note range. For example, if the connected keyboard has 76 keys, MorphMaster can adapt the horizontal axis of the Morphing Surface to that actual range.

In this way, the surface is effectively zoomed to the playable note range of the keyboard. This avoids wasting horizontal space on MIDI notes that cannot be played from the selected remote keyboard.

The same range is also used by the editors for **note morph curves**, because those curves operate on keyboard position.

Velocity morph curves, on the other hand, are driven by velocity and always use the standard MIDI velocity range **0–127**.

Play Mode

MorphMaster provides two playback modes:

- **Poly**
- **Mono**

Both modes are implemented internally by MorphMaster and do not rely on the playback modes provided by the target MIDI instrument.

For this reason, any instrument, synthesizer or sound module controlled by MorphMaster should normally be configured in **Poly** mode, even when MorphMaster itself is operating in Mono mode.

This approach guarantees consistent behaviour across different MIDI devices and avoids relying on device-specific implementations of polyphonic and monophonic playback.

In particular, MorphMaster's Mono mode provides a specialised Mono Retrigger behaviour that may not be available on all instruments, or may be implemented differently depending on the manufacturer. By handling note allocation internally, MorphMaster ensures identical results regardless of the connected MIDI device.

Poly

In Poly mode, MorphMaster operates as a normal polyphonic instrument. Any number of notes may be played simultaneously, limited only by the capabilities of the target MIDI device.

Each note is processed independently and contributes to the morphing engine according to its keyboard position, velocity, group assignments and morphing curves.

Mono

Mono mode implements a behaviour similar to the **Mono Retrigger** mode found on some synthesizers.

Only one note can sound at a time. Whenever a new note is played, any currently sounding note is immediately released and the newly played note becomes the active note.

However, previously pressed keys are not forgotten. MorphMaster keeps track of all notes that remain physically held down.

When the currently active note is released while other keys are still held, the most recently played of the remaining held notes becomes active again. In other words, the note is retriggered automatically when the note that was temporarily replacing it is released.

For example, if note C is played first and then note G is played while C is still held, G becomes the active note. When G is released, C is automatically retriggered and becomes the active note again.

The velocity used for the retriggered note is derived from the **Note Off velocity** of the note being released.

This behaviour was chosen because it provides direct expressive control over the dynamics of the retriggered note. The gesture that causes the previous note to return—releasing the currently active note—also determines how strongly the retriggered note will sound.

Presets

MorphMaster configurations can be stored as presets.

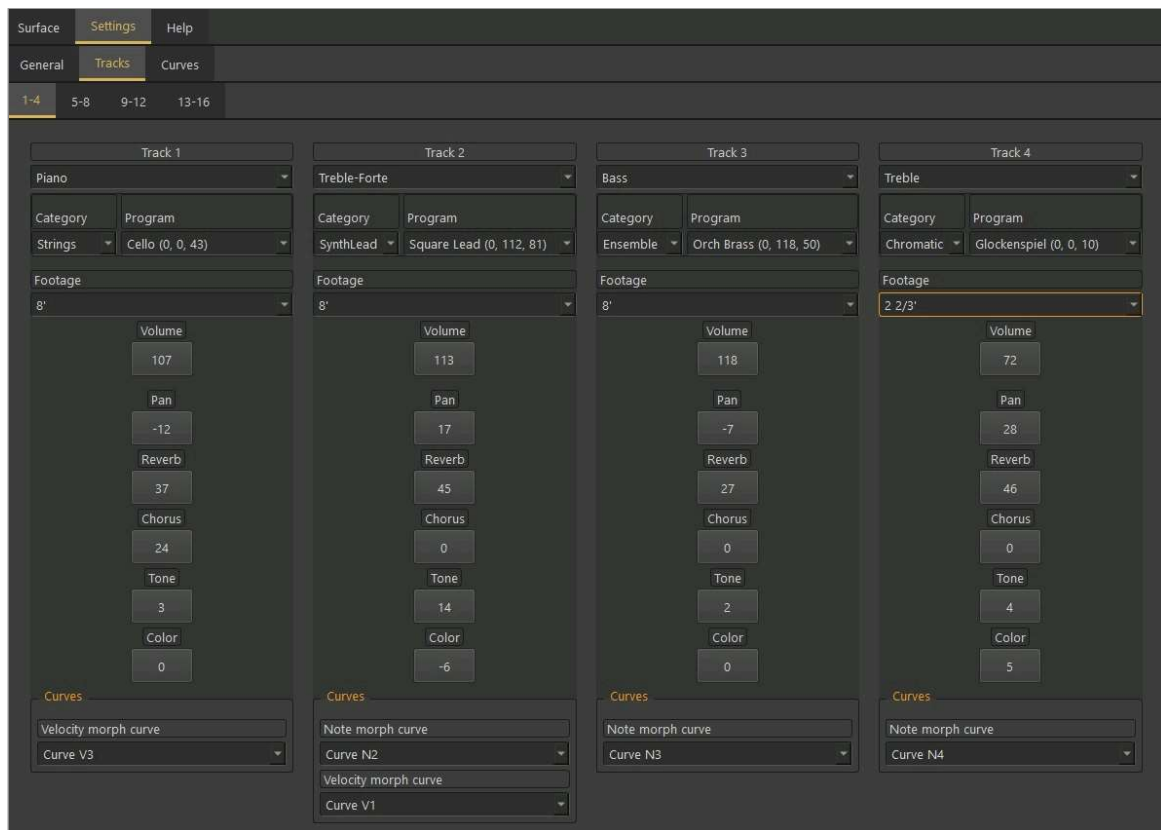
- **Save** updates the current preset.

- **Save As** creates a new preset.
- **Delete** removes the selected preset.

A preset stores:

- MIDI settings
 - Track assignments
 - Instrument selections
 - Remote keyboard range
 - Note and velocity morph curve definitions
 - Track parameters
-

3. Settings → Tracks



Tracks settings.

Track Configuration

MorphMaster can control up to sixteen MIDI tracks on the selected MIDI receiver device.

The actual sounds are provided by the remote instrument, synthesizer or sound module selected in the MIDI Output section. MorphMaster does not generate sounds itself; instead, it controls how the available receive tracks contribute to the final result.

All active tracks receive note events derived from the same control track. What distinguishes one track from another is not the notes being played, but the way each track participates in the morphing process. Different tracks may contain different sounds and may be assigned to different morphing groups.

The Tracks page is divided into four tabs:

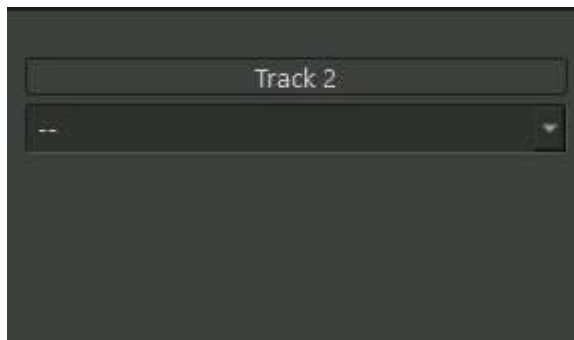
- Tracks 1–4
- Tracks 5–8
- Tracks 9–12
- Tracks 13–16

Group Assignment

A track only participates in the morphing process after it has been assigned to one of the eight morphing groups described earlier in this manual.

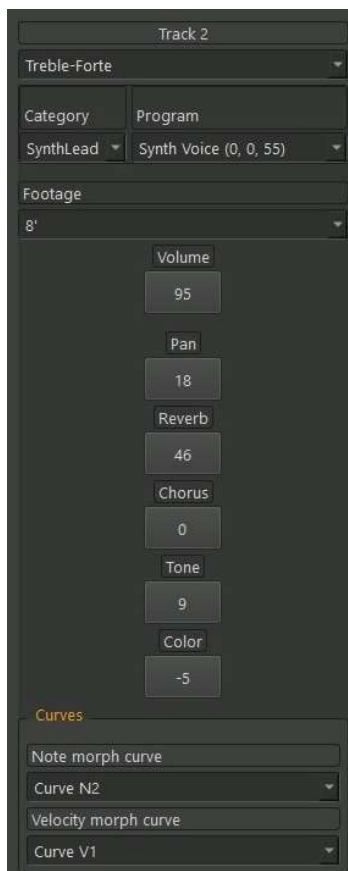
Tracks can be assigned to groups directly from the Morphing Surface by touching or clicking a group badge, or from the track strip associated with the track itself.

When a track is not assigned to any group, its track strip displays only the Group selector, with the value --.



Unassigned track strips.

Once a group has been selected, additional controls become available. These controls define how the track contributes to the final sound and how it interacts with the morphing engine.



Assigned track strip with all controls visible.

The selected group determines the region of the Morphing Surface in which the track is most active, while the remaining controls determine how its contribution evolves across keyboard position and playing velocity.

Instrument Selection

Each track can be configured to select a specific sound on the target MIDI receiver.

MorphMaster supports two different sound selection methods:

- **Instrument Definition**
- **Manual**

Instrument Definition Mode

Instrument Definition mode is the recommended method whenever the target instrument is included in MorphMaster's instrument database.

In this mode, sounds are selected using two selectors:

- Select a **Category**.
- Select a **Program** within that category.

The Program selector displays the sounds belonging to the selected category for the currently selected instrument. Program names appear exactly as they are defined on the target instrument, making sound selection easier and more intuitive.

MorphMaster automatically sends the required Bank Select and Program Change messages for the selected sound.



Instrument Definition mode.



Manual mode.

Manual Mode

Manual mode provides direct access to the MIDI patch parameters:

- Bank MSB
- Bank LSB
- Program Number

This mode is useful when controlling instruments that are not included in MorphMaster's instrument database, or when a specific patch must be selected using its raw MIDI bank and program numbers.

The Program selector contains the 128 standard General MIDI program names and numbers. These names are provided as a convenient reference, although the actual sound produced depends entirely on the target instrument and the selected bank values.

Footage Control

At first sight, the Footage control may appear unrelated to the morphing concept described in the previous sections. In reality, it is fully consistent with MorphMaster's goal of helping musicians explore sounds and timbral combinations that would be difficult or impossible to create on most MIDI keyboards.

The Footage control is inspired by the tradition of pipe organs and drawbar organs such as the Hammond. In these instruments, a complex sound is created by combining multiple components at different harmonic intervals.

In a pipe organ, the components are individual ranks of pipes. In a drawbar organ, they are generated by separate oscillators. In MorphMaster, the components are the individual tracks and the sounds assigned to them.

This makes it possible to build original timbres using an additive synthesis approach, where different programs are combined at harmonic intervals relative to a fundamental sound.

The available footage values are:

- 16'
- 8'
- $5 \frac{1}{3}'$
- 4'
- $2 \frac{2}{3}'$
- 2'
- $1 \frac{3}{5}'$
- $1 \frac{1}{3}'$
- $1 \frac{1}{7}'$
- 1'

Each footage value transposes the track so that it contributes a different harmonic component to the final sound.

For example, try combining:

- a Baritone Sax at 8'
- a Clarinet at $2 \frac{2}{3}'$
- a Flute at $1 \frac{3}{5}'$

The resulting timbre is often surprising and can be quite different from any of the individual sounds used to create it.

The real power of the Footage control emerges when it is combined with MorphMaster's morphing engine.

For example, imagine creating a custom additive timbre using the three tracks described above and assigning them to a Treble-oriented group. Then assign a Brass Ensemble sound to a Bass-oriented group.

When playing in the lower register, the Brass Ensemble will dominate the sound. As notes move towards the upper register, the contribution of the Brass Ensemble gradually decreases while the custom additive timbre becomes increasingly prominent.

The result is not merely a transition between two preset sounds, but a continuous evolution between fundamentally different sound structures. This combination of additive sound design and real-time morphing is one of the most distinctive features of MorphMaster.

Mixer Parameters

Each track provides a number of parameters that can be controlled directly from MorphMaster.

The complete sound setup remains the responsibility of the target MIDI receiver. Depending on the capabilities of the instrument, additional parameters may be available and can be configured directly on the instrument itself.

Only the parameters managed by MorphMaster are stored in MorphMaster presets. Settings modified directly on the remote instrument are not included in the preset data and therefore will not be restored when a preset is recalled.

The following parameters can be controlled and stored by MorphMaster:

- **Volume**
- **Pan**
- **Reverb**
- **Chorus**
- **Tone**
- **Color**

Volume

Controls the contribution of the track to the final mix.

Pan

Controls the stereo position of the track.

Reverb

Controls the amount of signal sent to the instrument's reverb effect.

Chorus

Controls the amount of signal sent to the instrument's chorus effect.

Tone

Controls the MIDI brightness parameter (CC 74).

Color

Controls the MIDI timbre parameter (CC 71).

Curve Assignment

MorphMaster provides two separate sets of editable morphing curves:

- **4 note morph curves**, used for morphings based on keyboard position.

- **4 velocity morph curves**, used for morphings based on velocity.

These curves define the crossfade profiles used by the tracks, that is, how gradually each track enters or leaves the final mix as the note position changes across the Morphing Surface.

Each active track must be assigned to one or two curves, depending on the morphing group to which the track belongs.

If a track is assigned to one of the side groups — **Bass** (♭), **Treble** (♭), **Forte** (f) or **Piano** (p) — the crossfade takes place along a single axis.

- For **Bass** and **Treble**, the crossfade is driven by keyboard position and uses one of the **note morph** curves.
- For **Piano** and **Forte**, the crossfade is driven by velocity and uses one of the **velocity morph**

If a track is assigned to one of the corner groups — **Bass Forte** (♭f), **Treble Forte** (♭f), **Treble Piano** (♭p) or **Bass Piano** (♭p) — the crossfade takes place along both axes.

In this case, MorphMaster takes into account both keyboard position and velocity. The keyboard-position component uses one of the note morph curves, while the velocity component uses one of the velocity morph curves.

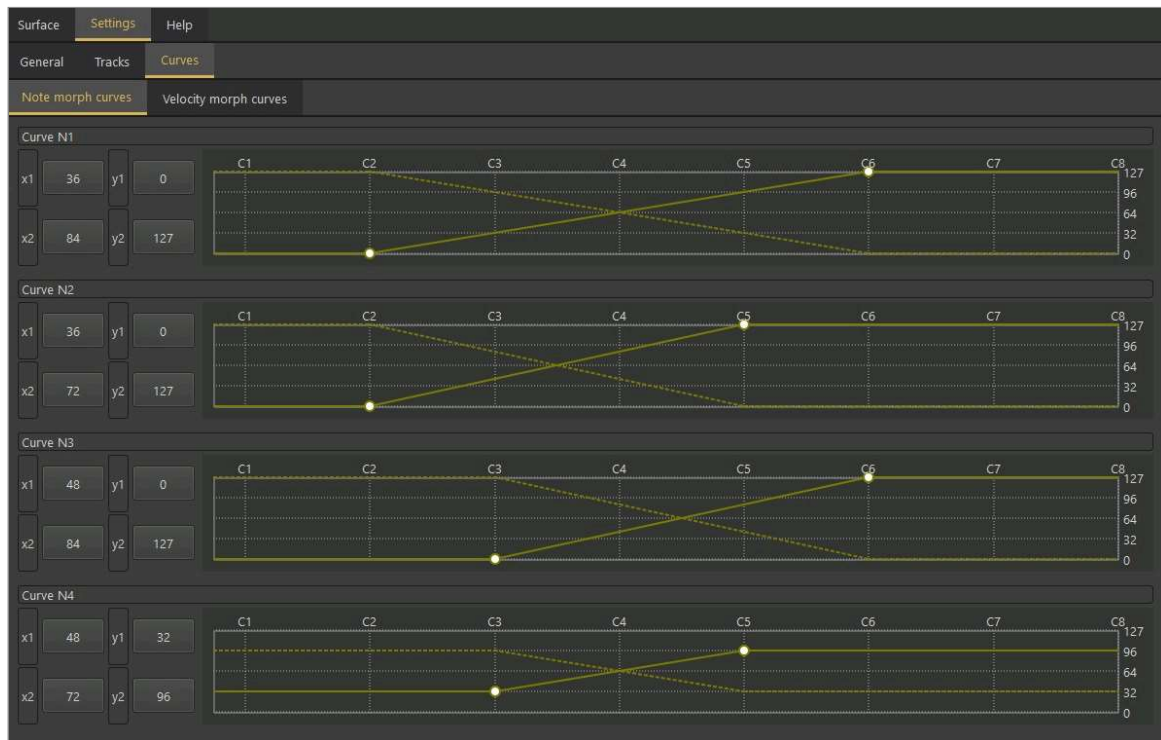
Keeping note morph and velocity morph curves separate makes the behaviour easier to understand. Note morph curves follow the selected Remote Keyboard Range, while velocity morph curves always use the MIDI velocity range 0–127.

In this way, the morphing behaviour is not fixed. Different tracks may use different curves, allowing some sounds to fade in very gradually while others appear more quickly.

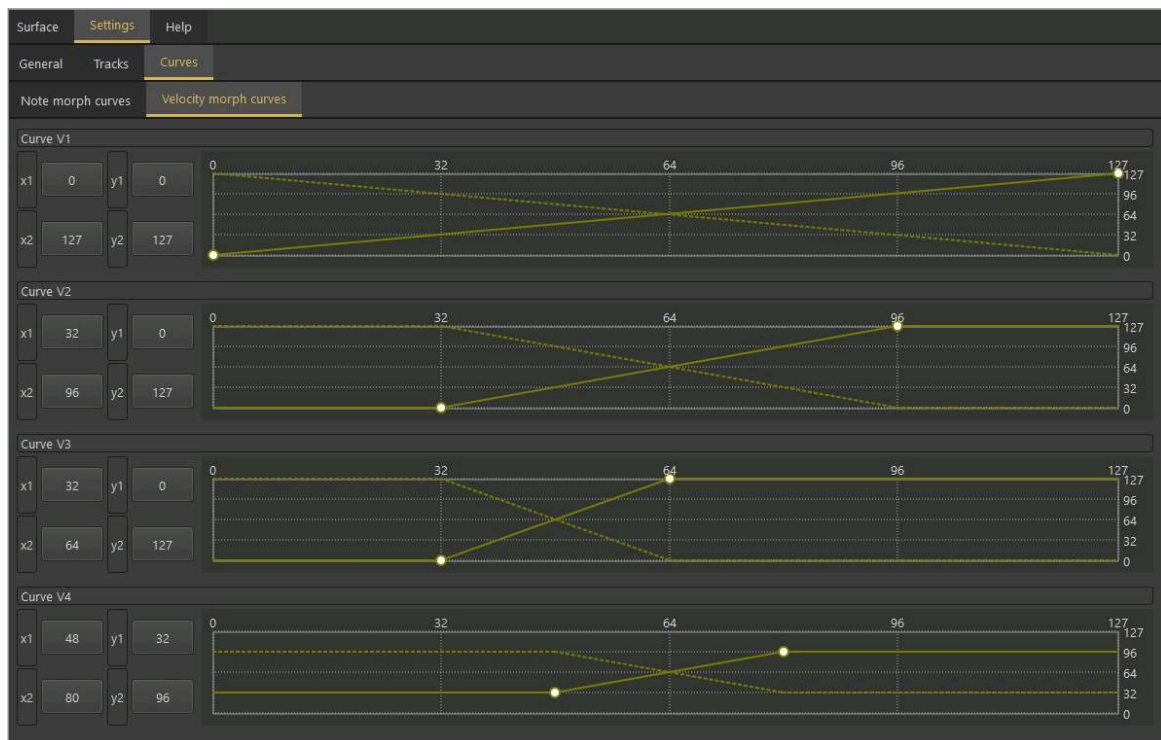
The note morph and velocity morph curves are stored as part of the current preset. This means that each preset can have its own completely different set of morphing curves.

The curves themselves, and the editor used to modify them, are described in detail in the **Settings → Curves** section.

4. Settings → Curves



Note morph curves settings.



Velocity morph curves settings.

Curve Concept

Morphing curves define how the contribution of a track changes while moving across the Morphing Surface.

MorphMaster provides two curve editors:

- **Note morph Curves**, used for morphings based on keyboard position.
- **Velocity morph Curves**, used for morphings based on velocity.

In both editors, the horizontal axis of the graph represents the input value used to control the crossfade.

For note morph curves, this input value is keyboard position. The displayed range follows the **Remote Keyboard Range** selected in **Settings → General**.

For velocity morph curves, the input value is MIDI velocity. The displayed range is always **0–127**.

The vertical axis of the graph represents the intensity scale factor applied to the track. Its range is **0–127**, where 0 means that the track is silent and 127 means that the track contributes at full intensity.

Curve Definition

For crossfading between tracks, MorphMaster uses curves that can be edited as three-segment responses:

- a first horizontal segment
- a central transition segment
- a final horizontal segment

This type of response can be defined by two control points: (x_1, y_1) and (x_2, y_2) .

The value x_1 is the keyboard-position or velocity value at which the transition begins, depending on whether the curve belongs to the note or velocity morph curve set.

The value x_2 is the keyboard-position or velocity value at which the transition ends. The value y_2 is the final intensity scale factor reached at the end of the transition.

Editing a curve therefore means deciding where the fade-in or fade-out should begin, where it should end, and which intensity values should be used before and after the transition.

Graph Editor

The graphical editor allows the two control points to be edited either numerically or directly on the graph.

The points can be dragged to change the region in which the crossfade takes place. Moving them horizontally changes the key or velocity values at which the transition begins and ends. Moving them vertically changes the initial and final intensity scale factors.

Solid and Dashed Curves

Each graph displays two curves at the same time.

The **solid curve** is the curve being edited directly. The **dashed curve** is its complementary curve, mirrored around the middle intensity value.

These two curves are shown together because a crossfade normally involves two complementary behaviours: while one track fades out, the other track fades in.

For example, suppose Track 1 is assigned to the **Piano** group and Track 2 is assigned to the **Forte** group. Since this crossfade is based on velocity, both tracks should use the same velocity morph curve, for example **Velocity morph Curve 1 (Curve V1)**.

MorphMaster will use one side of the curve pair to reduce the contribution of Track 1 as velocity increases, and the complementary side to increase the contribution of Track 2.

The same principle applies to keyboard-position crossfades. For example, if Track 1 is assigned to **Bass** (♭) and Track 2 is assigned to **Treble** (♯), both tracks should use the same note morph curve in order to obtain a complementary transition across the keyboard.

In this way, editing one curve automatically defines both directions of the crossfade. The solid and dashed curves are displayed together to make this complementary relationship visible.

Internal Crossfade Behaviour

The curves shown in the editor are displayed as simple linear segments for clarity and ease of editing.

Internally, MorphMaster uses a more refined crossfade function to preserve a balanced overall contribution between the tracks involved in the morphing process. This avoids unwanted level changes that could occur with a purely linear crossfade.

The visual representation remains valid for editing purposes: the two control points still define where the transition begins, where it ends, and the intensity values before and after the transition. Only the internal shape of the transition segment is refined by MorphMaster.

MIDI Messages Generated

MorphMaster generates standard MIDI messages:

- Note On
- Note Off
- Program Change
- Bank Select
- Control Change
- All Notes Off
- Reset Controllers

No proprietary MIDI protocol is required.

Version Information

MorphMaster Version 1.0

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